ISAIAH THOMPSON OCANSEY, Ph.D.

Postdoctoral Fellow · Machine Learning & Applied Research
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PROFESSIONAL SUMMARY

Applied machine learning researcher with 7+ years of experience developing predictive models, anomaly-detection methods, and analytics workflows across research, business, and quantitative-risk settings. Builds and evaluates models using Python and PyTorch, with experience in experimentation, model monitoring, distributed training, and cloud platforms. First author of LogTinyLLM, which reported 98.83% accuracy on log anomaly detection while updating fewer than 0.05% of model parameters.

TECHNICAL SKILLS

Machine Learning & AI: Predictive modeling, classification, deep learning, compact LLMs, transformers, NLP, anomaly detection, feature engineering, LoRA/parameter-efficient fine-tuning, federated learning, differential privacy, model compression

Programming & Data: Python, SQL, R, Bash, data mining, data cleaning, pandas, NumPy, scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face Transformers

Statistics & Experimentation: Statistical inference, hypothesis testing, A/B testing, experimental design, survival analysis, time series, statistical process control

ML Engineering & Infrastructure: Model monitoring, drift detection, distributed training, FSDP, mixed precision, Docker, Git, CI/CD, Linux, Slurm, GPU computing, HPC

Cloud & Analytics: AWS, Microsoft Azure, Azure ML, Apache Spark, REST APIs, ETL pipelines, Power BI, Tableau

PROFESSIONAL EXPERIENCE

O'Donnell Data Science and Research Computing Institute (ODSRCI), Southern Methodist University · Dallas, TX · Oct 2026 – Present
Postdoctoral Fellow, Artificial Intelligence and Machine Learning

- Conduct applied research at the intersection of AI/ML and high-performance computing, focusing on resource-efficient AI, scalable distributed computing, and edge systems.
- Design and evaluate machine learning approaches to estimate and optimize resource-efficiency parameters under complex, real-world conditions.
- Collaborate with interdisciplinary researchers on institute initiatives using NVIDIA SuperPOD and campus HPC infrastructure, and mentor students.

University of Texas at El Paso, Department of Mathematical Sciences · El Paso, TX · Jan 2022 – Dec 2025
Graduate Research Assistant, Machine Learning and Quantitative Modeling

- Developed LogTinyLLM using LoRA fine-tuning of compact language models for contextual log anomaly detection; reported 98.83% accuracy versus 79.37% for a fully fine-tuned LogBERT baseline while updating fewer than 0.05% of parameters. First-author paper under review at IEEE Transactions on Neural Networks and Learning Systems (TNNLS).
- Developed DP-FLogTinyLLM, a cross-silo federated learning framework with differential privacy that achieved 98.76% accuracy without centralizing client data. First-author paper under review.
- Implemented distributed PyTorch training with mixed precision and FSDP on TACC supercomputers; mentored three undergraduate researchers whose work was presented publicly.

University of Texas at El Paso, Graduate School · El Paso, TX · May 2023 – Aug 2024
Data Analytics Consultant, Strategic Decision Support

- Built student-success forecasting models with 94% accuracy and provided analysis for Board of Regents enrollment-policy discussions, contributing to a reported 15% institutional enrollment increase.
- Created self-service Power BI dashboards consolidating 10+ institutional KPIs, reducing ad hoc reporting by 40%.

University of Texas System · Austin, TX · Jun 2025 – Aug 2025
Information Technology Intern, Machine Learning and Data Engineering

- Integrated 15+ IT service management (ITSM) sources across 14 institutions through REST APIs and Power Automate into Power BI reporting, improving operational-risk visibility and reducing project risk exposure by 30%.
- Automated compliance reporting pipelines, saving 15+ hours of manual reporting each week and improving audit readiness by 25%.
- Researched federal AI funding opportunities and identified more than $2M in potential grant opportunities.

AT&T Inc. · Prague, Czech Republic · Aug 2018 – Dec 2021
Technical Specialist, Quantitative Risk and Data Analytics

- Led customer-experience analytics that identified systemic service failures, supporting $1.2M in annual revenue retention and a 20% increase in Net Promoter Score.
- Led a team of eight analysts applying statistical process control across 500+ cell sites; reduced service disruptions by 15% and improved uptime/latency metrics by 18%.

SELECTED PROJECTS

Drift Arena · github.com/ocansey/drift-arena

- Built a champion/challenger fraud-model monitoring workflow using Population Stability Index (PSI) drift detection and McNemar-test-gated promotion; reproduced recall degradation from 97% to 25% under an unseen attack profile.

The Label Budget · github.com/ocansey/label-budget

- Implemented Isolation Forest, HistGradientBoosting, and RBF-SVM approaches from scratch to study unsupervised-guided sampling, demonstrating a path to cut labeled-data requirements by half.

SELECTED PUBLICATIONS

- Ocansey et al. (2025). “LogTinyLLM: Tiny Large Language Models Based Contextual Log Anomaly Detection.” IEEE Transactions on Neural Networks and Learning Systems (under review). arXiv:2507.11071
- Ocansey et al. (2026). “DP-FLogTinyLLM: Differentially Private Federated Log Anomaly Detection Using Tiny LLMs.” Under review. arXiv:2604.19118

EDUCATION

University of Texas at El Paso · May 2026
Ph.D., Data Science
Dissertation: “Multi-Modal Anomaly Detection Using Deep Learning Techniques.” Advisor: Dr. Ritwik Bhattacharya.

University of Silesia, Poland · 2017
M.Sc., Mathematics of Finance and Economics

University of L’Aquila, Italy · 2017
M.Sc., Engineering Mathematics